AAYUSH ARYAN

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EXPERIENCE

Zopper — Associate Software Engineer *Jan 2026 – Present*

Backend SPOC, Equitas Bank · Backend / InsurTech

- **Led a cross-functional team** of backend and frontend developers, designers, and QA testers delivering the Equitas Bank insurance platform — owned requirements end-to-end and drove delivery from analysis through deployment and production support.
- Developed and integrated **100+ REST APIs** across **10+ insurer integrations** and **5+ modules** (Health, Life, Motor – 2W/4W/Commercial, Sachet), with insurer-specific business rules behind a unified, config-driven interface.
- Implemented **premium-calculation and policy-issuance logic for revenue-critical financial workflows**, using database transactions and idempotent processing to guarantee correctness and prevent duplicate charges or policies.
- Owned deployment and reliability: **Dockerized services across AWS and OCI**, CI/CD releases, Celery + Redis async processing, VAPT security remediation, DC/DR disaster-recovery drills, and production incident diagnosis.

PROJECTS

Back Office — Insurance Policy Data Platform *Django · DRF · MySQL · Celery · Redis*

- Sole backend engineer on the bank's insurance policy reporting platform — a Django system where a **nightly cron scans each insurer's SFTP folder overnight** to fetch **80 daily file feeds** (16 insurers × 5 report types), normalizing them into a unified 100-column canonical schema, enriching against **8 read-only production databases**, and serving filterable APIs, downloadable reports, and scheduled MIS email.
- Cut large filtered sales-report generation from **~30s to 2–3s (~90%)** by eliminating an N+1 of cross-database master-data lookups — replacing per-row resolution with set-based batched queries (one DISTINCT pass per report) plus memoized employee/product caches — and streaming exports via queryset.iterator() with write-only workbooks so memory stays flat at any row count.
- Built the **ENVI/Non-ENVI attribution engine** matching policies across 5 production databases on a different key per line of business at ~5 queries per 1,000 records, and a **declarative per-insurer ingestion layer** (16 column-mapping profiles, fuzzy folder resolution that refuses ambiguous matches) so onboarding an insurer is configuration, not code.
- Engineered the **nightly Celery pipeline** (ingest → commercials → employee master → classification → payment status → reverse-feed push-back → summary email) with isolated re-runnable stages and MD5-based idempotency, and enforced a **custom DB router making writes to all 8 external production databases structurally impossible**.

Health Monitoring & Platform-Adoption Module (Azure AD SSO) *Django · Celery · SAML 2.0 · MySQL*

- Built platform-adoption analytics and API health-monitoring modules across a **three-service Django architecture with no shared database** (**~7,700 LOC, 25 REST endpoints, 56 tests**); Celery-backed scheduled aggregation reconciles lead/quote/policy data into local daily-metrics tables, validated against **11,887 production quotes over 168 days**.
- Diagnosed and fixed a **70% over-count** in production lead metrics — upstream row-cloning on reassignment made 74,672 rows represent only 22,030 real leads; reproduced it against live records and replaced naive counting with a MIN(created_at) cohort rule.
- Designed a configuration-driven **API health-check framework with 5 execution modes**, including a passive TCP/TLS probe for endpoints whose real invocation issues live policies, and an async-poll mode yielding one health signal per insurer from a single polling loop.
- Implemented **Azure AD / Entra ID SAML 2.0 SSO (SP-initiated, MFA)** with a replay-protection ledger, host-header-independent destination validation, fail-closed identity resolution, and an HMAC blind index over a Fernet-encrypted email column; diagnosed and resolved a full-service outage caused by an xmlsec/lxml libxml2 mismatch failing at import.

SKILLS

Languages: C, C++, Python
Backend: Django, Django REST Framework (DRF), RESTful API design, Microservices architecture
Databases: MySQL, PostgreSQL, SQLite, MongoDB
Async & Infrastructure: Redis, RabbitMQ, Celery, Docker, Docker Compose, Nginx, Git, CI/CD, Linux, AWS, OCI
Security & Auth: JWT, OAuth 2.0, SAML 2.0 SSO, Azure AD / Entra ID, blind-index lookup, VAPT remediation, OWASP secure coding
Performance: SQL & query optimization, indexing, caching strategies, distributed-systems debugging

ACHIEVEMENTS

- **HackerRank / GDSC Coding Competition ’22** — Ranked 1st out of 450 candidates.
- **Nibble Computer Society Recruitment** — Secured 2nd position among 690 candidates in the coding round.
- **CodeChef Starters 108** — Ranked 6th globally.
- **Codeforces Round 926 (Div. 2)** — Ranked 2290 of 43k+ globally; reached 1399 rating.
- **LeetCode** — Knight badge (rating 1845).

EDUCATION

JSS Academy of Technical Education, Noida — B.Tech *2021 – 2025*

CGPA: 7.45